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V12Bio

Virudhunagar District
Common Half Yearly Examination - 2025

Standard 12
BIOLOGY
Bio-Botany
Section-A

Time: 3.00 Hours Marks: 70 Marks: 35

I. Answer all the questions: **8x1=8**

1) Arrange the layers of anther wall from locus to periphery?
   a) Epidermis middle layer, tapetum, endothecium
   b) Tapetum, middle layer, epidermis, endothecium
   c) Endothecium, epidermis, middle layer tapetum
   d) Tapetum, middle layer, endothecium, epidermis
2) An allohexaploidy contains
   a) six different genomes b) six copies of three different genomes
   c) Two copies of three different genomes d) six copies of one genome
3) Plasmids are
   a) circular protein molecule b) required by bacteria
   c) tiny bacteria d) confer resistance to antibiotics
4) Orobanche is a
   a) saprophyte b) Partial parasite c) total parasite d) Autotroph
5) Which of the following is not a sedimentary cycle?
   a) Nitrogon cycle b) phosphorous cycle c) sulphur cycle d) calcium cycle
6) ..... plant does not grow in temperature countries like canada and Germany
   a) Apple b) Mango c) Grapes d) pear
7) Match the list I with II

List I Biofertilizer	List II Organisms
i) Free living N_2	a) Aspergillus
ii) Symbiolic N_2	b) Amanita
iii) P solubilizing	c) Anabaena azollae
iv) P mobilizing	d) Azotobacter

   a) i-c, ii-a, iii-b, iv-d b) i-d, ii-c, iii-a, iv-b
   c) i-a, ii-c, iii-b, iv-d d) i-b, ii-a, iii-d, iv-c
8) Golden rice differs from its parental strain by the addition of ..... beta carotene
   a) 4 b) 2 c) 7 d) 3

Section-B

II. Answer any four questions: **4x2=8**

9) What is endothelium?
10) Write the disadvantages of Bt cotton
11) Write any four ecologically important days
12) Differentiate linkage and crossing over
13) What is Albedo effect and write their effects
14) Differentiate primary introduction from secondary introduction

Section-C

III. Answer any three questions: Q.No. 18 is compulsory. **3x3=9**

15) Draw the diagram PBR322 plasmid and label the parts
16) Write the advantages of artificial seeds
17) How do sacred groves help in the conservation of biodiversity?
18) Trichoderma - bio control agent - Justify this statement'
19) How is anemochory differ from zoochory

Section-D

IV. Answer all the questions: **2x5=10**

20) a) Enumerate the characteristic features of Entomophilous flower **(OR)**
    b) What is single cell protein. Mention its application
21) a) Write the aims of plant breeding **(OR)**
    b) Write the botanical name of state tree of TamilNadu
       i) From where it is originated
       ii) Write its three uses

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Bio-Zoology Marks: 35

I. Answer all the questions: **8x1=8**

1) In which type of parthenogenesis are only males produced.
   a) Arrhenotoky b) Thelytoky c) Amphitoky d) Both a and b
2) The mature sperms are stored in the
   a) seminiferous tubules b) vas deferens
   c) epididymis d) seminal vesicle
3) Pataus syndrome is also refered to as
   a) 13-Trisomy b) 18-Trisomy c) 21-Trisomy d) None of these
4) **Assertion (A)** : Ovulation is the release of ovum from the Graafian follicle
   Reason (R) : It occurs during the follicular phase of the menstrual cycle
   a) A and R are true. R is the correct explanation of A
   b) A and R are true. R is not the correct explanation of A
   c) A is true. R is false
   d) Both A and R are false
5) Choose the correctly match pair
   a) Amphetamines - stimulant
   b) LSD - Narcotic
   c) Heroin - Psychotropic
   d) Benzodiazepine - Pain killer
6) Cyclosporin - A is an immuno suppressive drug produced from .....
   a) Aspergillus niger b) Monascus purpureus
   c) Penicillium notatum d) Trichoderma polysporum
7) Interferans is .....
   a) Antibody b) Bacterial antibody
   c) fungal antibody d) viral antibody
8) Some animals became inactive during summer. It is an example for
   a) structural adaptation b) Behavioural adaptation
   c) Psychological adaptation d) Physiological adaptation

II. Answer any four of the questions: **4x2=8**

9) Differentiate External and Internal fertilization
10) In genetic code which is called as non-sense codon?
11) What are called connecting links? Give examples
12) Define antibiotics
13) What is pedogenesis?
14) What is Red data book

III. Answer any three of the questions: **3x3=9**

15) Differentiate Vasectomy and Tubectomy
16) What are the applications of Karyotyping?
17) What is vaccine? What are its types?
18) Differentiate between somatic cell gene therapy and germline therapy?
19) How does recycling help reduce pollution?

IV. Answer the following: **2x5=10**

20) a) Give a schematic representation of spermatogenesis in humans
   (OR)
   b) How is sex determined in human beings?
21) a) What are the salient features of HGP?
   (OR)
   b) List out the various causes for biodiversity losses?

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